

Ritesh Somashekar

Python/UI Developer

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PROFESSIONAL SUMMARY

- **Python/UI Developer** with **6+ years** of experience enhancing and modernizing data platforms, building **web dashboards and APIs**, **distributed data-processing pipelines**, and **containerized applications** using **Python, Java, SQL, FastAPI, and Apache Spark**. Proven experience in **access control and authorization (OAuth 2.0)**, **job scheduling**, and **workflow orchestration (Apache Airflow, 20+ production pipelines)**.
- Strong background in **profiling and optimizing long-running compute workloads** — cutting iteration cycles from **3 days to under 18 hours**, preprocessing overhead by **70%**, and release cycles from **2 weeks to 5 days**. Experienced with **Docker, Kubernetes**, and **CI/CD** on **AWS** and **Azure**, delivering **HIPAA-** and **PCI-DSS-compliant** systems.
- Demonstrated success designing **user-facing dashboards and interfaces** (FastAPI, Tableau, Power BI) with clear **information hierarchy, usability, and consistent layout**. Adept at **technical design discussions, code reviews, and technical documentation**, collaborating effectively across engineering, product, research, and client teams.

TECHNICAL SKILLS

Programming & Scripting: Python (pandas, NumPy, scikit-learn, PyTorch), Java, SQL (PostgreSQL, Oracle), R, Scala

Web UI, APIs & Security: FastAPI, RESTful Microservices, API Versioning, OAuth 2.0, Dashboard Design, Tableau, Power BI, Data Visualization

Job Scheduling & Orchestration: Apache Airflow (DAGs), HPC Job Scheduling (ORC Hopper), Kafka, AWS SQS, Event-Driven Workflows, CI/CD

Containerization & DevOps: Docker, Kubernetes, Jenkins, Git, DVC, MLflow

Distributed Processing & Performance: Apache Spark, PySpark, AWS EMR, Hadoop, Hive, Sqoop, ETL Pipelines, Delta Lake, Performance Tuning, Optuna

Cloud & Databases: AWS (S3, EC2, Lambda, Glue, RDS, Athena, SageMaker), Azure (Stream Analytics, Data Services), MongoDB, Snowflake, Databricks

ML & NLP: PyTorch, Core ML, TensorFlow, XGBoost, LightGBM, HuggingFace Transformers, BERT/BioBERT, Llama 3, SHAP

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA.

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India.

PROFESSIONAL EXPERIENCE

George Mason University | UI/UX Developer

Fairfax, VA | August 2024 - Present

- Designed and implemented a **multi-user research dashboard** backed by a **FastAPI** and **Docker** service, structuring **information hierarchy and layout** across analysis views and serving results to users with **sub-200ms latency**.
- Built **Tableau** visualizations and reports for disaster-response and social-cohesion research, improving **usability and consistency** of how findings were presented to research stakeholders.
- Scheduled and ran **long-running compute jobs** (graph analytics and ensemble forecasting) on the **ORC Hopper HPC cluster** using **Python** and **R**, achieving an **RMSE of 0.89** on large-scale real-world datasets.
- Analyzed and optimized **end-to-end processing pipelines** in **Python, PyTorch, and MLflow**, automating tuning with **Optuna** to cut iteration cycles from **3 days to under 18 hours**.
- Built a transformer-based **document-processing pipeline (BioBERT, HuggingFace)** over **30K documents**, improving entity-recognition F1-score by **18%** and standardizing **15GB** across 6 datasets with **30%** less preparation time.
- Engineered **distributed feature pipelines** on **Apache Spark** and **AWS EMR** for large-scale telemetry data with **privacy-preserving data compliance**, and served on-device models at **sub-150ms latency** for millions of daily requests.
- Streamlined deployment workflows with **MLflow** and **FastAPI**, reducing release cycles from **2 weeks to 5 days**, enabling **A/B testing**, and shrinking model size by **40%** (LoRA, QAT) within **3%** of baseline accuracy.
- Built a **LightGBM** anomaly-detection service with threshold-based alerting under **PCI-DSS** compliance, reducing false positives by **25%**, with **SHAP**-based explainability for governance reviews.

- Collaborated in **design discussions and code reviews** (Git, DVC) and authored **technical documentation** and reproducibility standards, reducing onboarding time for new contributors to **under 2 days**.

Carelon Global Solutions | Python Developer (Client: Accenture)

India | January 2019 – August 2023

- Developed **RESTful microservices** with **OAuth 2.0** authorization and **API versioning** for secure enterprise system communication, supporting **HIPAA-compliant** data exchange and real-time **EHR integration**.
- Designed and optimized **20+ data pipelines** for provider and consumer analytics, orchestrated with **Apache Airflow DAGs** and **CI/CD** on **AWS**, scheduling ingestion, feature engineering, and retraining jobs and reducing manual effort by **25%**.
- Streamlined **end-to-end ingestion workflows** by building **Python** automation for **NPPES** provider data, improving data accuracy by **74%** and reducing processing overhead in claims-balancing workflows.
- **Profiled and optimized** large-scale processing by migrating from **Hive to Apache Spark/PySpark** and re-engineering **SQL** feature pipelines, boosting model-training efficiency by **60%** and cutting preprocessing overhead by **70%**.
- Built scalable **distributed ETL pipelines** with **PySpark**, **AWS (S3, Glue, SageMaker)**, and **Delta Lake**, and web crawlers that structured **400,000+** XML/JSON documents, improving data reliability by **30%**.
- Containerized and deployed Python services using **Docker**, **Kubernetes**, and **Jenkins CI/CD**, improving system uptime by **20%**.
- Developed **Tableau** and **Power BI** dashboards and a real-time **Azure Stream Analytics** pipeline surfacing **performance metrics** and anomalies for IoT sensor data, reducing unplanned downtime by **20%**.
- Delivered **BERT/BioBERT** clinical entity extraction (**80% accuracy**, **55%** faster processing) and rule-based invoice classification that reduced manual error rates by **70%**; designed **MongoDB** models for high-throughput sensor data.
- Collaborated with cross-functional teams and **Accenture** client stakeholders on the **Seven Plus Locations** project through **technical design discussions and code reviews**, reducing project turnaround time by **15%**.